Task 1 — Temperature Monitoring & LCD Display

Objective

The objective of this task is to measure temperature using a digital temperature sensor and display the real-time value on a 16x2 I2C LCD. The temperature readings are also logged on the serial monitor for observation and verification.

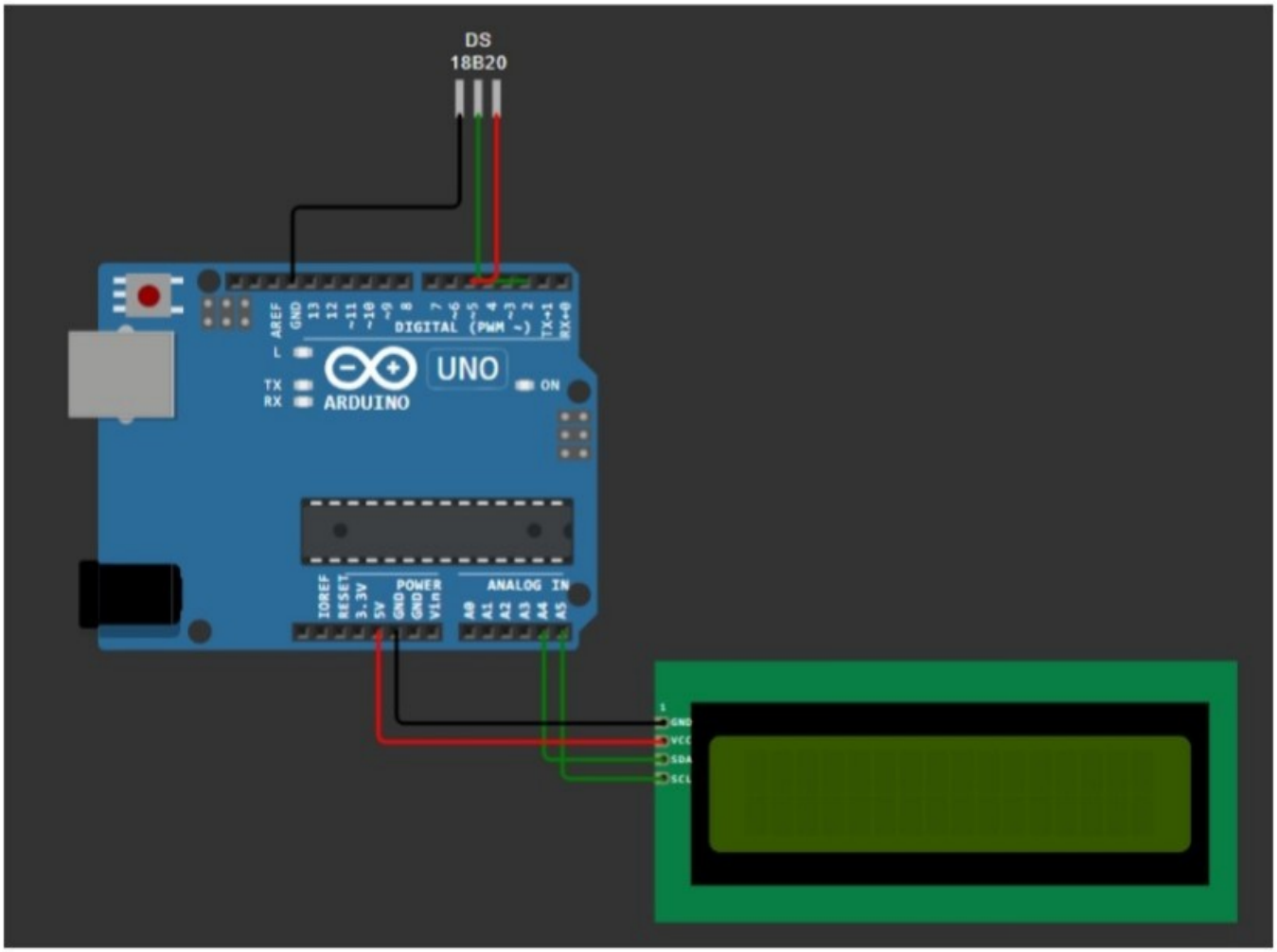
Components Used

- Arduino UNO
- DS18B20 Temperature Sensor
- 16x2 LCD Display (I2C Interface)
- Jumper Wires .

Working principle

The DS18B20 sensor measures the ambient temperature and sends the data to the Arduino UNO through the OneWire communication protocol. The Arduino processes the temperature value and displays the real-time reading on the 16x2 I2C LCD screen. The same temperature value is also printed on the Serial Monitor for logging and debugging purposes. The temperature updates continuously at a fixed interval.

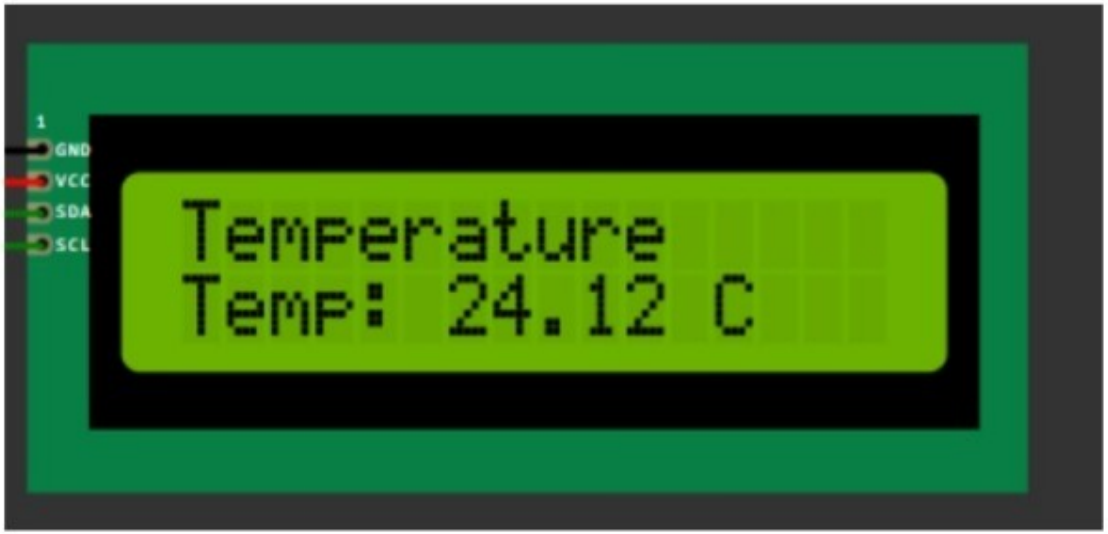
Circuit Diagram



Code

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include <LiquidCrystal_I2C.h>
#define ONE_WIRE_BUS 2
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup() {
  Serial.begin(9600);
  sensors.begin();
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("Temperature");
}
void loop() {
  sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);
  lcd.setCursor(0,1);
  lcd.print("Temp: ");
  lcd.print(tempC);
  lcd.print(" C ");
  Serial.print("Temperature: ");
  Serial.print(tempC);
  Serial.println(" C");
  delay(2000);
}
```

Output



Demo

New Arduino Uno Project - Wokwi Simul

Result

The temperature sensor successfully measured the environmental temperature and displayed the real-time value on the 16x2 LCD. The readings were also logged on the Serial Monitor, confirming proper working of the circuit and code.